

B. Sc. (Hons) Semester IV Open Book Examination 2020 – 21**CHEMISTRY****Paper No. CHB-401 : Inorganic Chemistry–II and Organic Chemistry–III****Time : 4 hours 30 minutes****Full Marks : 70****Instructions :**

- (i) The Question Paper contains 08 questions out of which you are required to answer any 2 questions each from Sections A and B. The question paper is of 70 Marks with each question carrying 17 · 5 Marks.
- (ii) The total duration of the examination will be 4:30 Hours (Four Hours and Thirty Minutes), which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
- (iii) For the students with benchmark disability as per Persons with Disability Act, the total duration of examination shall be 6 Hours (Six Hours) to complete the examination process, which includes the time for downloading the question paper from the portal, writing the answers by hand and uploading the hand-written answer sheets on the portal.
- (iv) Answers should be hand-written on plain white A4 size paper using black or blue pen. Each question can be answered in up to 350 words on 3 (Three) plain A4 size paper (only one side to be used).
- (v) Answers to each question should start from a fresh page. All pages are required to be numbered. You should write your Course Name, Semester, Examination Roll Number, Paper Code, Paper Title, Date and Time of Examination on the first sheet used for answers.
- (vi) Use separate Answer-Sheets for each Section.

Section—A
(Inorganic Chemistry–II)
(Marks : 35)

- Q.1.**
- 1. What happens when CH_3COOH is dissolved in liq. NH_3 , water and H_2SO_4 ? Explain its nature in above solvents by using Brønsted-Lowry Theory. 4½
 - 2. An aqueous solution of the compound $\text{CoN}_5\text{H}_{15}\text{Cl}_3$ gives a negative test for NH_3 and Cl^- and the solution is non-conducting. Write formula, hybridization and structure of the compound. **3**
 - 3. Calculate Effective Atomic Numbers (EAN) of the metal in the following : **3**
 - (i) $[\text{Ni}(\text{NH}_3)_6]^{2+}$
 - (ii) $\text{Mn}_2(\text{CO})_{10}$
 - (iii) $\text{K}_4[\text{Fe}(\text{CN})_6]$
 - 4. Why $[\text{Mn}(\text{H}_2\text{O})_6]\text{Cl}_2$ is very light in colour but $[\text{MnO}_4]^-$ is intensely coloured? **3**

5. Mention two lanthanide elements that show other stable oxidation states together with +3 and account for their stability. 4
- Q.2.**
1. Arrange the order of acidity of boron halides BF_3 , BCl_3 , BBr_3 and BI_3 in decreasing acid strength with proper explanations. 4
 2. Identify the Brønsted-Lowry acids and bases and their conjugate counter parts in the following : 6
 - (i) $\text{NH}_4^+ + \text{S}^{2-} \rightleftharpoons \text{HS}^- + \text{NH}_3$
 - (ii) $\text{CO}_3^{2-} + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{OH}^-$
 - (iii) $\text{H}_2\text{SO}_4 + \text{HClO}_4 \rightleftharpoons \text{H}_3\text{SO}_4^+ + \text{ClO}_4^-$
 3. Complete the following reactions and identify Lux-Flood acids and bases in each : 4
 - (i) $\text{K}_2\text{O} + \text{SO}_3 \longrightarrow$
 - (ii) $\text{Al}_2\text{O}_3 + \text{Na}_2\text{O} \longrightarrow$
 4. What is HSAB principle? How is this principle followed by nature in occurrence of metals as ores in different forms? $3\frac{1}{2}$
- Q.3.**
1. Write IUPAC names of the following : 6
 - (i) $\text{K}_2[\text{Zn}(\text{NH}_2)_4]$
 - (ii) $[(\text{NH}_3)_4\text{Cr}-\text{OH}-\text{Cr}(\text{NH}_3)_4]\text{Cl}_5$
 - (iii) $[\text{Co}(\text{en})_2\text{Cl}(\text{NO}_2)]\text{Cl}$
 - (iv) $[\text{Cr}(\text{NH}_3)_6][\text{Co}(\text{CN})_6]$
 2. On the basis of VBT, comment on hybridization, geometry and spin magnetic moment of the following : 6
 - (i) $[\text{Fe}(\text{CN})_6]^{3-}$
 - (ii) $[\text{Cu}(\text{NH}_3)_4]^{2+}$
 - (iii) $[\text{Cr}(\text{NH}_3)_6]^{3+}$
 3. $[\text{Ni}(\text{CO})_4]$ and $[\text{Ni}(\text{CN})_4]^{2-}$ are tetrahedral but significantly differ in their magnetic behaviour. Explain it by using Valence Bond Theory. $2\frac{1}{2}$
 4. Draw and indicate the type of possible isomers of the following : 3
 - (i) $[\text{Cr}(\text{NH}_3)_4(\text{Co}(\text{CN})_6)]$
 - (ii) $[\text{Ru}(\text{O})_2\text{Cl}_2(\text{en})]$
 - (iii) $[\text{Cr}(\text{oxalate})_3]^{3-}$
- Q.4.**
1. Complete the following reactions in liquid ammonia : 4
 - (i) $(\text{NH}_4)_2\text{S} + \text{CuCl}_2 \longrightarrow$
 - (ii) $\text{Ba}(\text{NO}_3)_2 + 2\text{AgBr} \longrightarrow$
 - (iii) $\text{BiN} + 3\text{NH}_4\text{NO}_3 \longrightarrow$
 - (iv) $\text{NaNH}_2 + \text{NH}_4\text{I} \longrightarrow$

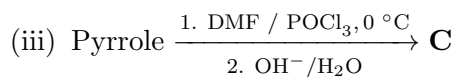
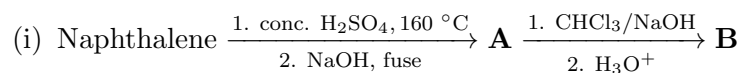
2. What is the effect of variation of concentration of alkali metal solution in liq. NH_3 on colour and conductivity of the solution? **3**
 3. Write ground state electronic configuration of Tm^{2+} , Dy^{4+} and Nd^{3+} and indicate the number of unpaired electrons in each case. **$4\frac{1}{2}$**
 4. How is $(\text{CH}_3)_3\text{Sn}(\text{C}_2\text{H}_5)$ synthesized in liq. NH_3 ? Compare and explain the acidic behaviour of $[\text{NH}_2]_2\text{SO}_2(\text{OH})$ in water and liq. NH_3 . **6**
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Section—B
(Organic Chemistry—III)
(Marks : 35)

- Q.1.** 1. Discuss the mechanism and orientation of electrophilic substitution reaction in pyrrole. **$3\frac{1}{2}$**
2. Write a chemical synthesis for indigotin starting from anthranilic acid. **3**
3. Write the chemical structures of α and β D (+) glucose. **2**
4. Give a reaction to confirm that glucose and fructose have same configuration at C-3, C-4 and C-5. **3**
5. Complete the following reactions : **3×2**
- (i) (*Naphthalene functional ring execution matrices*) $\xrightarrow[2. \text{KOH} / \Delta]{1. \text{Cl-CH}_2\text{CH}_2\text{COCl} / \text{AlCl}_3} ?$
- (ii) (*Phenanthrene core conversions*) $\xrightarrow{\text{HNO}_3 + \text{H}_2\text{SO}_4} ? \xrightarrow{[\text{O}]} ?$
- (iii) (*Furan layout transformations*) $\xrightarrow[\Delta]{\text{CaO}} ?$
- Q.2.** 1. How will you establish the structure of indigotin? **$5\frac{1}{2}$**
2. How will you synthesize the following? **3×2**
- (i) 4-Nitropyridine from pyridine
- (ii) 2-Furoic acid from furan
- (iii) Fluorescein from resorcinol
3. Complete the following reactions : **2×3**
- (i) Benzaldehyde + N,N-Dimethylaniline $\xrightarrow[100^\circ\text{C}]{\text{Conc. H}_2\text{SO}_4} ? \xrightarrow{\text{PbO}_2 / \text{AcOH}} ? \xrightarrow{\text{H}^+} ?$
- (ii) Fructose $\xrightarrow{\text{Conc. H}_2\text{SO}_4} ?$
- Q.3.** 1. Give a reaction sequence to convert : **2×3**
- (i) Glucose into fructose
- (ii) D-arabinose into D-glucose
2. Despite being a ketohexose, why fructose is a reducing sugar? Explain. **$2\frac{1}{2}$**

3. Why do malachite green and fluorescein develop color in solution? Explain. **3**

4. Predict the products in the following reactions : **2×3**



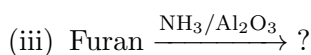
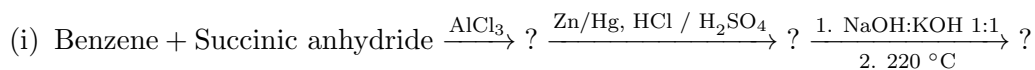
Q.4. 1. Explain the order of aromaticity in pyrrole, furan and thiophene. **2½**

2. Write the stepwise mechanism for the following reactions : **2×3**

(i) Wohl's degradation method

(ii) Feist-Benary synthesis

3. Complete the following reactions : **3+2+2**



External Portals & Academic Resources:

- **Examination Portal:** <https://bhu-exam.vercel.app/>
- **Student Resource Center:** <https://quashan-me.vercel.app/>
- **Academic Portfolio:** <https://me-aryan.vercel.app>